

CSS @font-face Rule: Complete Guide

CSS Learning Resources

May 9, 2026

Contents

1	Introduction	2
1.1	Why @font-face Matters	2
2	Basic @font-face Syntax	2
2.1	Core Properties	2
3	Font Formats and Browser Support	3
3.1	Format Comparison Table	3
3.2	Recommended Format Stack	3
4	Font-Display Property	3
4.1	Font-Display Values	4
4.2	Font-Display in Practice	4
5	Real-World Examples	4
5.1	Example 1: Google Fonts Integration	4
5.2	Example 2: Self-Hosted Custom Font	5
5.3	Example 3: Variable Fonts	5
5.4	Example 4: Subsetting for Performance	6
5.5	Example 5: OpenType Features	6
6	Performance Optimization	7
6.1	Font Loading Strategy	7
6.2	File Size Optimization	7
7	Browser Support	8
7.1	Current Support Status	8
8	Common Mistakes	8
8.1	Mistake 1: Wrong Font Path	8
8.2	Mistake 2: Forgetting Fallback Fonts	8
8.3	Mistake 3: Not Using font-display	9
8.4	Mistake 4: Mismatched Weight/Style Selectors	9
8.5	Mistake 5: Not Considering Performance Impact	9

9	Advanced Techniques	10
9.1	Font Preloading	10
9.2	Local Font Fallback	10
9.3	Font Subsetting by Language	10
9.4	Fallback Font Stack Strategy	11
10	Accessibility Considerations	11
10.1	Dyslexia-Friendly Fonts	11
10.2	High Contrast Requirements	12
10.3	Respect User Preferences	12
11	Integration with Font Services	13
11.1	Google Fonts	13
11.2	Typekit (Adobe)	13
11.3	Self-Hosted with CDN	13
12	Quick Reference	13
12.1	Minimal Setup	13
12.2	Complete Setup	13
12.3	Performance Checklist	14
13	Summary	14

1 Introduction

The `@font-face` at-rule is one of the most powerful features in modern CSS, enabling developers to use custom fonts beyond the limited set of system fonts. Introduced with CSS3, `@font-face` revolutionized web typography by allowing designers to implement any typeface in their projects, transforming web design from text-only constraints to limitless creative possibilities.

Before `@font-face`, web designers were restricted to a small handful of “web-safe” fonts that were guaranteed to exist on most user machines. This meant Arial, Georgia, Verdana, and a few others dominated the web. Today, custom web fonts have become essential to modern web design, enabling brands to maintain their visual identity across all digital platforms while improving accessibility through thoughtfully selected typefaces.

1.1 Why @font-face Matters

- **Brand Consistency:** Maintain your exact brand typography across all devices
- **Visual Impact:** Create distinctive designs that stand out in competitive markets
- **Accessibility:** Choose typefaces optimized for readability and dyslexia-friendly designs
- **Performance Control:** Optimize font loading for faster page rendering
- **Variable Fonts:** Support weight and style variations without loading multiple files

2 Basic @font-face Syntax

The fundamental structure of `@font-face` is straightforward. It declares a font family name and specifies where the browser should load the font file from.

```
1 @font-face {  
2     font-family: 'CustomFontName';  
3     src: url('path/to/font.woff2') format('woff2'),  
4         url('path/to/font.woff') format('woff');  
5     font-weight: normal;  
6     font-style: normal;  
7 }
```

Once defined, use the font family in your CSS:

```
1 body {  
2     font-family: 'CustomFontName', Arial, sans-serif;  
3 }
```

2.1 Core Properties

- **font-family:** Custom name for your web font (quoted or unquoted)
- **src:** URL and format of the font file(s)
- **font-weight:** 100 to 900 (normal, bold, etc.)
- **font-style:** normal, italic, or oblique

- **font-stretch**: Condensed, normal, expanded (for variable fonts)
- **unicode-range**: Subset of characters to load
- **font-display**: Controls how the font loads (auto, block, swap, fallback, optional)
- **font-variant**: OpenType features like small-caps, ligatures

3 Font Formats and Browser Support

Different font formats offer varying levels of browser support and compression. Understanding which formats to serve ensures your fonts display correctly across all devices.

3.1 Format Comparison Table

Format	Browser Support	Compression	Use Case
WOFF2	Modern browsers	Excellent (best)	Primary choice
WOFF	All modern, IE9+	Good	Fallback
TTF	Most browsers	Moderate	Legacy support
OTF	Most browsers	Moderate	OpenType features
EOT	IE6-8 only	Good	Ancient IE support
SVG	Legacy iOS/Android	Poor	Older mobile devices

3.2 Recommended Format Stack

```
1 @font-face {  
2     font-family: 'RobotoCustom';  
3     src: url('roboto.woff2') format('woff2'),  
4         url('roboto.woff') format('woff'),  
5         url('roboto.ttf') format('truetype');  
6 }
```

Modern approach (optimal):

```
1 @font-face {  
2     font-family: 'RobotoModern';  
3     src: url('roboto.woff2') format('woff2');  
4 }
```

4 Font-Display Property

The **font-display** property controls how the browser renders text while the font is loading. This is critical for user experience and perceived performance.

4.1 Font-Display Values

Value	Behavior	Best For
auto	Browser default (usually block)	Headings, body text
block	Hide text during load	Critical fonts
swap	Show fallback, swap when ready	Headlines
fallback	Brief delay, then fallback	Body text
optional	May not load at all	Nice-to-have fonts

4.2 Font-Display in Practice

```
1  /* Headlines: Show fallback immediately */
2  @font-face {
3      font-family: 'PrimaryHeading';
4      src: url('heading-font.woff2') format('woff2');
5      font-display: swap;
6  }
7
8  /* Body text: Slight delay, acceptable fallback */
9  @font-face {
10     font-family: 'BodyFont';
11     src: url('body-font.woff2') format('woff2');
12     font-display: fallback;
13 }
14
15 /* Decorative: Non-critical, optional loading */
16 @font-face {
17     font-family: 'DecorativeFont';
18     src: url('decorative.woff2') format('woff2');
19     font-display: optional;
20 }
```

5 Real-World Examples

5.1 Example 1: Google Fonts Integration

Using Google Fonts (simplest approach):

```
1  /* Link in HTML */
2  <link href="https://fonts.googleapis.com/css2?family=Roboto:wght@400;700&
3      display=swap" rel="stylesheet">
4
5  /* Use in CSS */
6  body {
7      font-family: 'Roboto', sans-serif;
8  }
9
10 h1 {
11     font-weight: 700;
12 }
```

5.2 Example 2: Self-Hosted Custom Font

Implementing a purchased or custom font:

```
1 @font-face {
2     font-family: 'Montserrat';
3     src: url('/fonts/Montserrat-Regular.woff2') format('woff2'),
4         url('/fonts/Montserrat-Regular.woff') format('woff');
5     font-weight: 400;
6     font-style: normal;
7     font-display: swap;
8 }
9
10 @font-face {
11     font-family: 'Montserrat';
12     src: url('/fonts/Montserrat-Bold.woff2') format('woff2'),
13         url('/fonts/Montserrat-Bold.woff') format('woff');
14     font-weight: 700;
15     font-style: normal;
16     font-display: swap;
17 }
18
19 @font-face {
20     font-family: 'Montserrat';
21     src: url('/fonts/Montserrat-Italic.woff2') format('woff2'),
22         url('/fonts/Montserrat-Italic.woff') format('woff');
23     font-weight: 400;
24     font-style: italic;
25     font-display: swap;
26 }
27
28 h1, h2, h3 {
29     font-family: 'Montserrat', sans-serif;
30     font-weight: 700;
31 }
32
33 body {
34     font-family: 'Montserrat', sans-serif;
35     font-weight: 400;
36 }
37
38 em, i {
39     font-style: italic;
40 }
```

5.3 Example 3: Variable Fonts

Variable fonts support multiple weights in a single file:

```
1 @font-face {
2     font-family: 'InterVariable';
3     src: url('/fonts/Inter-variable.woff2') format('woff2-variations');
4     font-weight: 100 900;
5     font-display: swap;
```

```
6 }
7
8 body {
9     font-family: 'InterVariable', sans-serif;
10    font-weight: 400;
11 }
12
13 strong {
14     font-weight: 700;
15 }
16
17 .light-text {
18     font-weight: 300;
19 }
20
21 .extra-bold {
22     font-weight: 900;
23 }
```

5.4 Example 4: Subsetting for Performance

Load only specific characters (e.g., Latin subset):

```
1 @font-face {
2     font-family: 'RobotoSubset';
3     src: url('roboto-latin.woff2') format('woff2');
4     unicode-range: U+0000-00FF, U+0131, U+0152-0153;
5     font-display: swap;
6 }
7
8 body {
9     font-family: 'RobotoSubset', sans-serif;
10 }
```

5.5 Example 5: OpenType Features

Enable advanced typography features:

```
1 @font-face {
2     font-family: 'TypographyPro';
3     src: url('typography-pro.woff2') format('woff2');
4     font-feature-settings: 'liga' 1, 'calt' 1;
5 }
6
7 /* Elegant numerals and ligatures */
8 .elegant-text {
9     font-family: 'TypographyPro', serif;
10    font-variant-numeric: oldstyle-nums;
11    font-feature-settings: 'liga' 1, 'dlig' 1;
12 }
13
14 /* Small caps */
15 .scaps {
```

```
16     font-variant-caps: small-caps;
17 }
```

6 Performance Optimization

6.1 Font Loading Strategy

```
1  /* 1. Critical fonts: Block render until loaded */
2  @font-face {
3      font-family: 'CriticalFont';
4      src: url('critical.woff2') format('woff2');
5      font-display: block;
6  }
7
8  /* 2. Important fonts: Swap as soon as available */
9  @font-face {
10     font-family: 'HeadingFont';
11     src: url('heading.woff2') format('woff2');
12     font-display: swap;
13 }
14
15 /* 3. Body fonts: Brief wait, acceptable fallback */
16 @font-face {
17     font-family: 'BodyFont';
18     src: url('body.woff2') format('woff2');
19     font-display: fallback;
20 }
21
22 /* 4. Decorative fonts: Optional, don't block */
23 @font-face {
24     font-family: 'DecorativeFont';
25     src: url('decorative.woff2') format('woff2');
26     font-display: optional;
27 }
```

6.2 File Size Optimization

```
1  /* Subset Latin extended: Reduce from 200KB to 50KB */
2  @font-face {
3      font-family: 'SubsetRoboto';
4      src: url('roboto-subset.woff2') format('woff2');
5      unicode-range: U+0000-024F;
6      font-display: swap;
7  }
8
9  /* Preload critical fonts in HTML */
10 <link rel="preload" as="font" type="font/woff2"
11     href="/fonts/roboto.woff2" crossorigin>
12
13 /* Font self-hosting */
14 @font-face {
```



```
15 font-family: 'LocalFont';
16 src: local('Roboto'),
17      url('roboto.woff2') format('woff2');
18 font-display: swap;
19 }
```

7 Browser Support

7.1 Current Support Status

Feature	Chrome	Firefox	Safari	Edge	IE
@font-face	All	All	All	All	9+
WOFF2	36+	39+	12+	15+	No
font-display	60+	58+	11+	79+	No
Variable fonts	62+	62+	11+	79+	No
OpenType features	All	All	All	All	Limited

8 Common Mistakes

8.1 Mistake 1: Wrong Font Path

Wrong:

```
1 @font-face {
2   font-family: 'MyFont';
3   src: url('MyFont.woff2') format('woff2');
4 }
```

Correct:

```
1 @font-face {
2   font-family: 'MyFont';
3   src: url('/fonts/MyFont.woff2') format('woff2');
4 }
```

8.2 Mistake 2: Forgetting Fallback Fonts

Wrong:

```
1 body {
2   font-family: 'CustomFont';
3 }
```

Correct:

```
1 body {
2   font-family: 'CustomFont', Georgia, serif;
3 }
```

8.3 Mistake 3: Not Using font-display

Missing optimization:

```
1 @font-face {  
2     font-family: 'SlowFont';  
3     src: url('font.woff2') format('woff2');  
4 }
```

Optimized:

```
1 @font-face {  
2     font-family: 'OptimizedFont';  
3     src: url('font.woff2') format('woff2');  
4     font-display: swap;  
5 }
```

8.4 Mistake 4: Mismatched Weight/Style Selectors

Wrong:

```
1 @font-face {  
2     font-family: 'MyFont';  
3     src: url('MyFont-Bold.woff2') format('woff2');  
4     font-weight: 400;  
5 }  
6  
7 strong {  
8     font-weight: 700;  
9 }
```

Correct:

```
1 @font-face {  
2     font-family: 'MyFont';  
3     src: url('MyFont-Regular.woff2') format('woff2');  
4     font-weight: 400;  
5 }  
6  
7 @font-face {  
8     font-family: 'MyFont';  
9     src: url('MyFont-Bold.woff2') format('woff2');  
10    font-weight: 700;  
11 }  
12  
13 strong {  
14     font-weight: 700;  
15 }
```

8.5 Mistake 5: Not Considering Performance Impact

Inefficient:

```
1 @font-face {  
2     font-family: 'DecorativeFont';
```

```
3     src: url('decorative.woff2') format('woff2');
4     font-display: block;
5 }
6
7 .fancy-button {
8     font-family: 'DecorativeFont', Arial, sans-serif;
9 }
```

Optimized:

```
1 @font-face {
2     font-family: 'DecorativeFont';
3     src: url('decorative.woff2') format('woff2');
4     font-display: optional;
5 }
6
7 .fancy-button {
8     font-family: 'DecorativeFont', Arial, sans-serif;
9 }
```

9 Advanced Techniques

9.1 Font Preloading

Preload critical fonts to reduce render-blocking time:

```
1 <head>
2     <link rel="preload" as="font" type="font/woff2"
3         href="/fonts/primary.woff2" crossorigin>
4     <link rel="preload" as="font" type="font/woff2"
5         href="/fonts/secondary.woff2" crossorigin>
6 </head>
```

9.2 Local Font Fallback

Use locally installed fonts first, then fall back to web fonts:

```
1 @font-face {
2     font-family: 'EffectiveFont';
3     src: local('Roboto'),
4         local('Roboto-Regular'),
5         url('/fonts/roboto.woff2') format('woff2');
6     font-display: swap;
7 }
```

9.3 Font Subsetting by Language

Serve only fonts needed for specific languages:

```
1 /* Latin alphabet */
2 @font-face {
3     font-family: 'UniversalFont';
4     src: url('font-latin.woff2') format('woff2');
```

```
5     unicode-range: U+0000-024F;
6 }
7
8 /* Greek alphabet */
9 @font-face {
10     font-family: 'UniversalFont';
11     src: url('font-greek.woff2') format('woff2');
12     unicode-range: U+0370-03FF;
13 }
14
15 /* Cyrillic alphabet */
16 @font-face {
17     font-family: 'UniversalFont';
18     src: url('font-cyrillic.woff2') format('woff2');
19     unicode-range: U+0400-04FF;
20 }
```

9.4 Fallback Font Stack Strategy

```
1 /* Serif: Professional, traditional */
2 body {
3     font-family: 'CustomSerif', 'Georgia', 'Times New Roman', serif;
4 }
5
6 /* Sans-serif: Modern, clean */
7 h1 {
8     font-family: 'CustomSans', 'Helvetica', 'Arial', sans-serif;
9 }
10
11 /* Monospace: Code, technical */
12 code {
13     font-family: 'CustomMono', 'Courier New', monospace;
14 }
15
16 /* System font stack: Fast, fallback */
17 .system-optimized {
18     font-family: -apple-system, BlinkMacSystemFont,
19                 'Segoe UI', 'Helvetica Neue', sans-serif;
20 }
```

10 Accessibility Considerations

10.1 Dyslexia-Friendly Fonts

Choose fonts optimized for readability:

```
1 @font-face {
2     font-family: 'AccessibleDyslexia';
3     src: url('dyslexia-font.woff2') format('woff2');
4     font-display: swap;
5 }
6
```

```
7 body {  
8     font-family: 'AccessibleDyslexia', sans-serif;  
9     line-height: 1.6;  
10    letter-spacing: 0.05em;  
11 }
```

10.2 High Contrast Requirements

Ensure sufficient contrast with background:

```
1 @font-face {  
2     font-family: 'AccessibleFont';  
3     src: url('accessible.woff2') format('woff2');  
4 }  
5  
6 body {  
7     font-family: 'AccessibleFont', sans-serif;  
8     color: #000;  
9     background: #fff;  
10 }  
11  
12 /* High contrast mode */  
13 @media (prefers-contrast: more) {  
14     body {  
15         color: #000;  
16         background: #fff;  
17         font-weight: 600;  
18     }  
19 }
```

10.3 Respect User Preferences

Honor system font preferences:

```
1 @media (prefers-reduced-motion: reduce) {  
2     @font-face {  
3         font-family: 'OptimizedFont';  
4         src: url('optimized.woff2') format('woff2');  
5         font-display: block;  
6     }  
7 }  
8  
9 @media (prefers-color-scheme: dark) {  
10     body {  
11         font-family: 'CustomFont', sans-serif;  
12         color: #e0e0e0;  
13     }  
14 }
```

11 Integration with Font Services

11.1 Google Fonts

```
1 @import url('https://fonts.googleapis.com/css2?family=Roboto:wght@300;400;700&
  display=swap');
2
3 body {
4     font-family: 'Roboto', sans-serif;
5 }
```

11.2 Typekit (Adobe)

```
1 <link rel="stylesheet" href="https://use.typekit.net/abc1234.css">
```

```
1 body {
2     font-family: 'adobe-typekit-font', sans-serif;
3 }
```

11.3 Self-Hosted with CDN

```
1 @font-face {
2     font-family: 'CDNFont';
3     src: url('https://cdn.example.com/fonts/font.woff2') format('woff2');
4     font-display: swap;
5 }
```

12 Quick Reference

12.1 Minimal Setup

```
1 @font-face {
2     font-family: 'CustomFont';
3     src: url('font.woff2') format('woff2');
4     font-display: swap;
5 }
6
7 body {
8     font-family: 'CustomFont', fallback-font, sans-serif;
9 }
```

12.2 Complete Setup

```
1 @font-face {
2     font-family: 'RobotoComplete';
3     src: url('/fonts/roboto-regular.woff2') format('woff2'),
4         url('/fonts/roboto-regular.woff') format('woff');
```

```
5     font-weight: 400;
6     font-style: normal;
7     font-display: swap;
8 }
9
10 @font-face {
11     font-family: 'RobotoComplete';
12     src: url('/fonts/roboto-bold.woff2') format('woff2'),
13         url('/fonts/roboto-bold.woff') format('woff');
14     font-weight: 700;
15     font-style: normal;
16     font-display: swap;
17 }
18
19 body {
20     font-family: 'RobotoComplete', Arial, sans-serif;
21     font-weight: 400;
22 }
23
24 strong, b {
25     font-weight: 700;
26 }
```

12.3 Performance Checklist

- Use WOFF2 format for modern browsers
- Add font-display: swap to all @font-face rules
- Preload critical fonts with <link rel="preload">
- Subset fonts when possible to reduce file size
- Use variable fonts to reduce file count
- Provide fallback system fonts
- Host fonts on CDN for faster delivery
- Monitor font loading performance with Web Vitals
- Test on slow 4G networks
- Consider lazy-loading decorative fonts

13 Summary

The **@font-face** rule is fundamental to modern web design, enabling custom typography that enhances user experience and brand identity. Key takeaways:

1. Use WOFF2 format for optimal compression and browser support
2. Always include font-display for performance optimization

3. Match font-weight and font-style selectors to actual font files
4. Provide fallback fonts for graceful degradation
5. Preload critical fonts to reduce render-blocking
6. Consider performance impact on loading time and Core Web Vitals
7. Subset fonts by language and characters when appropriate
8. Respect accessibility requirements and user preferences
9. Use variable fonts when possible to reduce file transfers
10. Test fonts across different devices and network conditions

Master @font-face and you unlock unlimited typographic possibilities for your web projects.